

Department of Computer Science



Submitted in part fulfilment for the degree of BSc Computer Science
with Artificial Intelligence

Comparative Machine Learning Techniques for Predicting Macroeconomic Indicators

[Tunahan Sisman](#)

April 2025

Supervisor: [Dr. Poonam Yadav](#)

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Dr. Poonam Yadav, for her continuous guidance, support and encouragement throughout this project. Her insightful feedback and expertise have been instrumental in shaping both the direction and quality of this work.

STATEMENT OF ETHICS

This project did not involve any human participants, personal data, biological material, or interaction with vulnerable populations, and therefore did not require formal ethical approval from the Physical Sciences Ethics Committee (PSEC). All data used in this study were sourced from the UK Office for National Statistics (ONS) and are publicly available under the Open Government Licence (OGL) v3.0. These datasets include national-level macroeconomic indicators such as GDP, unemployment rate, inflation rate, current account balance, and trade data (imports and exports), and do not contain any personally identifiable or sensitive information.

In line with the University's ethical principles of avoidance of harm, informed consent, and data protection, the project posed no ethical risks to individuals, animals, the environment, or broader communities. No dual-use applications, military contexts, or collaborations with controversial sponsors were involved. The datasets used did not require additional permission, nor were they subject to any restrictions on dissemination. The entire research process was conducted within the boundaries of academic integrity and responsible data usage.

All models and techniques applied—whether statistical or machine learning-based—were selected based on their academic relevance and evaluated objectively. The outcomes of this study are intended solely for academic purposes, with no commercial, military, or ethically sensitive applications.

This ethical position has been reviewed and confirmed in consultation with the project supervisor, Dr. Poonam Yadav, and is compliant with the University of York's Code of Practice and Principles for Good Ethical Governance.

Table of Contents

Executive Summary.....	vii
1 Introduction.....	1
2 Background and Literature Review	2
2.1 Key Macroeconomic Indicators in Forecasting.....	2
2.2 Traditional Macroeconomic Forecasting Methods	3
2.2.1 ARIMA.....	3
2.2.2 VAR.....	4
2.2.3 GARCH.....	5
2.2.4 Other Traditional Forecasting Models.....	6
2.3 ML Macroeconomic Forecasting Methods	6
2.3.1 Supervised Learning Models	6
2.3.2 Unsupervised Learning Models	7
2.3.3 Hybrid and Ensemble Learning Models	8
2.4 Comparing ML and Traditional Forecasting Models.....	10
2.4.1 Advantages of ML Models.....	10
2.4.2 Limitations of ML Models	10
2.5 Forecast Evaluation Metrics and Validation Techniques.....	11
3 Methodology	14
3.1 Research Approach	14
3.2 Selected Models.....	14
3.3 Workflow	14
3.3.1 Data Acquisition and Preprocessing	15
3.3.2 Model Implementation.....	15
3.3.3 Performance Evaluation	15
3.4 Data Processing Approach.....	16
3.4.1 Handling Missing Values	16
3.4.2 Data Normalization and Transformation.....	16
4 Implementation	17
4.1 Data Sources.....	17
4.2 Macroeconomic Indicators as Predictors	17
4.2.1 Global Economic Factors.....	17
4.2.2 Local Economic Factors.....	17
4.3 Models Implementation	18
4.4 Tools and Programming Languages	18
4.5 Individual Model Implementation.....	18
4.5.1 VAR Model.....	18
4.5.2 ARIMA Model	20
4.5.3 LSTM Model.....	21
4.5.4 XGBoost Model	22

5 Results and Evaluation	24
5.1 VAR Model Evaluation	24
5.2 ARIMA Model Evaluation	25
5.3 LSTM Model Evaluation.....	25
5.4 XGBoost Model Evaluation.....	26
5.5 Models Comparison	27
6 Conclusion	29
Future Work	30
Appendices.....	31
Appendix A: ADF Test Results After Differencing (VAR).....	31
Appendix B: ARIMA Model Selection Based on BIC	31
Appendix C: VAR Lag Selection Grid Search	31
Appendix D: XGB Model Configuration.....	32
Appendix E: LSTM Model Structure and Training	32
Bibliography	33

List of Tables

Table 1: ADF Test Results Before Differencing.....	19
Table 2: ADF Test Results for GDP	20
Table 3: Top 3 ARIMA Models by AIC	20
Table 4: Evaluation metrics for each model on the dataset.	27
Table 5: ADF Test Results After Differencing	31
Table 6: Top 3 ARIMA Models by BIC	31

List of Figures

Figure 1: ACF of GDP Residuals	19
Figure 2: Time Series Plot of GDP Residuals from the VAR(2) Model	20
Figure 3: Residual Plot of ARIMA(3,0,0).....	21
Figure 4: ACF and PACF of Residuals for ARIMA(3,0,0)	21
Figure 5: LSTM Training and Validation Loss Across Epochs	22
Figure 6: ACF and PACF of LSTM Residual	22
Figure 7: Top 15 Feature Importances in XGBoost GDP Forecasting	23
Figure 8: Var Model Forecast vs Actual GDP	24
Figure 9: ARIMA Model Forecast vs Actual GDP	25
Figure 10: LSTM Model Forecast vs Actual GDP	26
Figure 11: XGBoost Forecast vs Actual GDP	27
Figure 12: GDP Growth Rate: Actual vs Models Forecasts	28
Figure 13: Grid Search Results for VAR Model.....	31
Figure 14: Final Hyperparameter configuration used for the XGB Model.....	32
Figure 15: Architecture and Training Configuration of the LSTM Model	32

Executive Summary

This project investigates the forecasting performance of both traditional statistical and modern machine learning techniques in predicting the United Kingdom's Gross Domestic Product (GDP) growth rate. As GDP remains a fundamental macroeconomic indicator guiding public policy, investment decisions, and strategic planning, enhancing its forecast accuracy is a critical pursuit. The overarching aim of this work is to determine whether machine learning methods—when applied with appropriate preprocessing and feature engineering—can outperform established econometric models in forecasting short-term GDP trends.

The motivation stems from the increasing complexity and volatility of global economic systems. Traditional models like ARIMA (Autoregressive Integrated Moving Average) and VAR (Vector Autoregression) are well-established in economic research for their interpretability and mathematical rigor but often struggle with capturing sudden nonlinear changes and interactions between variables. In contrast, machine learning approaches such as LSTM (Long Short-Term Memory) networks and XGBoost (Extreme Gradient Boosting) have gained attention for their flexibility, capacity to learn from data-driven features, and ability to model hidden nonlinear patterns without the need for strict assumptions.

To test these approaches, a unified macroeconomic dataset was constructed using publicly available quarterly UK data, spanning multiple decades. Key indicators such as inflation, unemployment, trade balance, and the current account were included, with GDP as the target variable. The data underwent preprocessing steps including differencing, normalization (using MinMaxScaler), and extensive feature engineering. Four forecasting models—ARIMA, VAR, LSTM, and XGBoost—were trained and evaluated using a consistent methodology, including a 70%-15%-15% train-validation-test split for machine learning models, hyperparameter tuning where applicable, and model evaluation using standard error metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2).

The results showed a clear performance gradient. XGBoost achieved the strongest results, with the lowest MAE and RMSE and the highest R^2 score, confirming its strength in leveraging structured economic features and modelling nonlinear dependencies. LSTM outperformed both traditional models by generating smoother forecasts and capturing temporal sequences, though its effectiveness was somewhat limited by the dataset's size and its computational

demands. ARIMA provided reasonably stable forecasts but lacked adaptability to dynamic trends. VAR, while capable of modelling multivariate interactions, struggled with overfitting and sensitivity to lag selection, ultimately resulting in the weakest performance across all metrics.

These findings affirm the hypothesis that machine learning models—particularly ensemble-based methods like XGBoost—can offer substantial improvements in macroeconomic forecasting accuracy when supported by robust data preparation and systematic evaluation. Importantly, the project also highlights the limitations of each approach and underscores the need for careful model selection and validation in economic applications.

The outcomes of this project have important implications for both academic research and practical forecasting applications. Firstly, they reinforce the growing consensus that machine learning models can complement—and in many cases, outperform—traditional econometric approaches in capturing complex economic dynamics. Secondly, the results establish a solid basis for future exploration into more advanced modelling strategies, including multivariate deep learning architectures, ensemble techniques that integrate multiple model types, and interpretability tools such as SHAP (SHapley Additive exPlanations) to enhance transparency and trust in model outputs. Taken together, this work offers a comprehensive and comparative evaluation of forecasting techniques, contributing valuable insights into the evolving role of artificial intelligence in macroeconomic prediction.

1 Introduction

In today's fast-paced global economy, accurate macroeconomic forecasting plays a crucial role in guiding policy, business decisions, and investment strategies. Economic indicators such as GDP, inflation, and employment offer essential insights into the future performance of national economies, helping stakeholders mitigate risk and respond proactively. As economies become increasingly data-rich and interconnected, the ability to extract meaningful forecasts from complex signals becomes both more feasible and more necessary.

However, forecasting economic trends is inherently challenging due to the complexity and volatility of financial systems. Numerous factors—including market fluctuations, geopolitical events, and technological advancements—can disrupt traditional economic patterns, making accurate predictions more difficult. In particular, unexpected global crises can introduce severe economic shocks that existing forecasting models struggle to capture.

This study investigates the comparative forecasting capabilities of four models—Autoregressive Integrated Moving Average (ARIMA), Vector Autoregression (VAR), Long Short-Term Memory (LSTM), and Extreme Gradient Boosting (XGBoost)—applied to the task of predicting quarterly UK GDP growth. While many prior studies have evaluated these models independently, this project uniquely integrates them within a single, systematic experimental pipeline, using a harmonized dataset and consistent evaluation criteria. The novelty lies in the direct, side-by-side comparison across classical and machine learning models, combined with an exploration of how preprocessing strategies such as differencing, scaling, and feature engineering affect forecasting accuracy.

Given the varying nature of macroeconomic indicators—some being smooth, others volatile—this research aims to identify which models best capture these dynamics. The findings aim to contribute to a deeper understanding of model performance in economic forecasting and offer insights for more resilient, data-driven decision-making in future policy and financial planning.

The remainder of this report is structured as follows: Section 2 presents a detailed literature review; Section 3 outlines the methodology; Section 4 covers implementation; Section 5 reports the results; and Section 6 provides conclusions and future works.

2 Background and Literature Review

This section provides a comprehensive review of the existing literature on macroeconomic forecasting and provides background on the key terminologies and techniques involved. It begins by introducing the concept and importance of economic forecasting, followed by a discussion of key macroeconomic indicators. The review then explores traditional statistical methods, modern machine learning techniques, and hybrid approaches used for forecasting. Finally, it compares the strengths and limitations of these models and outlines the most commonly used evaluation metrics.

Economic forecasting is the prediction of future scenarios of a country's economic activity and provides an overall outlook on variables such as economic growth, unemployment, and inflation among other examples [1]. It is key for policy makers as it serves as a solid foundation for planning across different scenarios by governments [2]. Historically, economic forecasting gained significant interest from scholars and academics starting with the Great Depression, as policymakers sought tools to anticipate and mitigate the likelihood of economic downturns [3].

Given the complexity of modern economies, producing accurate estimates has become increasingly important. For this, policy makers, academics and scholars use a range of models that vary from simple trend extrapolations to advanced statistical systems to project these indicators [4], [5], [6]. Notably, machine learning (ML) models have emerged in recent decades as a powerful complement, or even an alternative, to traditional forecasting methods [7], with the goal of enhancing prediction accuracy and capturing complex patterns within datasets [8].

Accordingly, the rest of the literature review will explore macroeconomic forecasting models. It will cover fundamental concepts, outline the most common traditional and ML techniques used in the field, and provide a comparison between the two categories. The objective is to synthesize theoretical frameworks and empirical evidence on how ML techniques have been applied in forecasting, and to set the foundation for the remaining sections of the study.

2.1 Key Macroeconomic Indicators in Forecasting

The existing literature shows that the forecast of macroeconomic indicators is centred on a few pivotal indicators that are considered by economists to be the most critical gauges of economic health. The first primary and most recurring metric forecasted in the literature review is the GDP [9]. Examples of scholars who have forecasted the GDP are

Modis [10] in the context of the United States (US), Tkacz [11] in the context of Canada, and Mumbuli et al. [12] in the context of Tanzania. This indicator measures the total monetary value of final goods and services produced within a period of time for a specific economy [13]. Therefore, forecasting this variable in terms of monetary value or in terms of percentage growth rate is considered as the most critical indicator that informs decisions on fiscal planning, government expenditure, spending, investment, and debt management among many other decisions [14], [15]. For example, if the GDP growth forecasts are positive and high, it might inform governments that the economy is expanding, and tax revenues are likely to increase. In response, the government may decide to increase public spending or reduce taxes to stimulate further economic growth.

Another macroeconomic indicator that is widely forecasted is inflation (e.g., [16], [17]). This indicator gives an overall idea about increases in the general price level of goods and services and is measured by various indices such as the Consumer Price Index (CPI) [18]. For example, high levels of the expected inflation might indicate that the economy is overheating. In other words, high inflation might be a signal that demand is outpacing supply. In this case, policy makers can increase taxes to decrease aggregate demand and stabilize prices.

Other key macroeconomic indicators that are widely forecasted in the existing literature are, but are not limited to, unemployment (e.g., [19], [20]), interest rates (e.g., [21]), exchange rates (e.g., [22], [23]), and wages among many other indicators. As discussed in the previous paragraphs, accurately forecasting these indicators is essential to support timely economic planning. In other words, accurate forecasts provide valuable insights that support governments, central banks, and the private sector to make informed decisions that will support economic resilience, stability, or growth [24]. However, failing to accurately forecast these indicators might lead to misguided policy choices that might lead to economic instability [25].

2.2 Traditional Macroeconomic Forecasting Methods

Traditional forecasting methods such as Autoregressive Integrated Moving Average (ARIMA) and Vector Autoregression (VAR) have been widely used in the literature to predict macroeconomic indicators. These models rely on historical patterns and statistical relationships to generate short-to medium- term forecasts.

2.2.1 ARIMA

The first and most common traditional forecasting model is the ARIMA model and its variants. The ARIMA model is data-driven and only relies on historical data and historical patterns in a single variable to make

predictions [26]. The strengths of these models lie in their ease of deployment, their reliance on one single variable, and in their accuracy in making short-term forecasts [27], [28]. Additionally, the ARIMA Model also accounts for trends, cycles, and seasonality by tuning its three main parameters (p,d,q) [29]. Mathematically, the general equation for the ARIMA model can be stated as [29]:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$

where:

- Y_t is the forecasted value at time t,
- p is the autoregressive order,
- d is the number of differencing steps to make the series stationary,
- q is the moving average order,
- ϕ_i and θ_j are model parameters,
- ε_t is the error term.

However, this model has its own limitations. First, the ARIMA model assumes linearity and struggles with nonlinear patterns. Second, the model does not account for sudden structural changes. Third, the model is not designed for making long-term forecasts [30].

The model has been massively used in the existing literature to predict macroeconomic indicators. In the contribution of Vafin [31], the author used ARIMA to forecast both unemployment and inflation in the context of US, United Kingdom (UK), China, France, Japan, Russia, and India. But for Kuzu and Alp [32], the authors used ARIMA to predict money supply, exchange rates, and interest rates for Turkey. For Uddin and Tanzim [33], the authors forecasted GDP in the context of Bangladesh. While the list of the papers that used ARIMA to forecast macroeconomic indicators is massive, other notable examples include the works of Alomani et al. [34], Advani et al. [35], and Ma [36].

2.2.2 VAR

The second most common traditional forecasting model is the VAR model. This technique might include multiple variables to forecast an indicator. Additionally, VAR models each variable as a linear function of its own lags and lags of other variables. Hence, the method accounts for feedback effects. Unlike ARIMA, VAR leverages information from multiple indicators, making it suitable for medium-term forecasts with reasonable accuracy. This model can mathematically be expressed using the following equation:

$$Y_t = c + A_1 Y_{t-1} + A_2 Y_{t-2} + \dots + A_p Y_{t-p} + \varepsilon_t$$

Where:

- Y_t is the $K \times 1$ vector of K endogenous variables at time t
- c is $K \times 1$ vector of intercepts
- A_i is $K \times K$ coefficient matrices for lag i
- p is the number of lags
- ε_t is $K \times 1$ vector of white noise error terms

However, it is also important to note the drawbacks of this method, as it also assumes linearity and struggles with nonlinear time series. Also, the VAR requires careful handling when data are non-stationary or when there is cointegration or long-run relationships between the variables included in the model. Finally, feeding the model with too many parameters might lead to overfitting, resulting to inaccurate or misleading forecasts.

In the contribution of Fokin and Polbin [37], the authors forecasted Russia's key macroeconomic indicators using the VAR method, and D'Agostino et al. [38] did a similar study in the context of the US. Similar to the ARIMA, this method is also widely used in predicting economic indicators.

2.2.3 GARCH

The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is widely used in financial and macroeconomic forecasting to model-time varying volatility [39]. Unlike ARIMA and VAR, which focus on predicting the expected future values, GARCH captures fluctuations around the mean, making it particularly suitable for forecasting volatility in variables like inflation and exchange rates.

The GARCH model consists of two components: a mean equation, which models the central tendency of a time series, and a variance equation, which models how volatility evolves over time based on past shocks and past volatility levels [40]. This structure helps explain not just what level a variable might reach, but also how uncertain or unstable that level may be.

Empirical studies demonstrate the model's relevance: Barbara et al. [41] used GARCH to forecast Ghana's GDP volatility, Abreu et al. [42] applied it to exchange rate forecasting, and Uwilingiyimana et al. [43] studied inflation volatility in Kenya.

2.2.4 Other Traditional Forecasting Models

In addition to the three models discussed above, it is important to note that there are several other traditional techniques that have been widely used in forecasting macroeconomic indicators. These include:

- The Holt-Winters models, used by Omane-Adjepong et al. [44] to forecast Ghana's inflation.
- The Phillips Curve, used by Gabrielyan [45] to forecast inflation in Sweden, Canada, and New Zealand.
- The Bayesian Vector Autoregression, used by Iyer [46] to forecast economic growth in India.

Each of these models has its own strengths and limitations. For instance, Holt-Winters models are used mainly to capture trends and seasonality in short-term forecasts [47], [48]. But for the Phillip Curve technique, it provides a theoretically grounded relationship between variables (e.g., inflation and unemployment), but its predictive accuracy is low [49].

Recognizing these limitations, the next section explores how ML models have emerged to address some of these challenges and enhance the forecasting of macroeconomic indicators.

2.3 ML Macroeconomic Forecasting Methods

ML offers a variety of techniques that have been increasingly applied in macroeconomic indicators forecasting. These techniques can be categorized into supervised, unsupervised, and hybrid/ensemble learning methods. Each of these categories contains models with unique advantages depending on the nature of the data as well as the objective of forecasting.

2.3.1 Supervised Learning Models

Supervised learning techniques are the most frequently used ML models for economic forecasting. These models are trained on historical data with known outcomes, allowing them to learn relationships between predictor variables (e.g., interest rates, industrial production, money supply) and target macroeconomic indicators [50], [51].

Among the simplest supervised models is linear regression, often used with regularization techniques such as Ridge or Lasso to prevent overfitting when dealing with high-dimensional data [52], [53]. However, the real strength of ML in this context lies in more advanced nonlinear methods. Decision trees and support vector regression (SVR) have all been tested in forecasting environments with promising results [54].

Of particular interest in recent literature is XGBoost (Extreme Gradient Boosting), a powerful tree-based algorithm that has gained popularity for its high accuracy, computational efficiency, and ability to handle noisy or incomplete datasets [55]. XGBoost builds a sequence of models, where each new model corrects the errors of its predecessor [56]. Its built-in regularization makes it especially suitable for macroeconomic datasets, which often involve a limited number of observations relative to the number of features. Empirical applications have demonstrated its effectiveness in forecasting economic indicators, especially when compared to traditional models [57].

Another major class of supervised models includes neural networks, especially deep learning architectures such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks [58], [59]. These are designed to process sequential data, making them ideal for time-series forecasting tasks. LSTM models have been shown to effectively capture long-term dependencies and lagged effects in macroeconomic indicators, outperforming traditional methods in many empirical studies. Their flexibility makes them suitable for capturing structural shifts and irregular patterns in economic cycles [60].

Recent studies confirm that supervised machine learning models have become increasingly prominent in forecasting macroeconomic indicators such as GDP, inflation, and human capital investment. For example, Zhangyin [61] developed a forecasting model using supervised ML techniques to predict inflation trends, demonstrating improved accuracy and adaptability compared to traditional methods. Similarly, Gupta et al. [62] emphasized the growing reliance on models like decision trees and SVM for capturing the nonlinear dynamics in economic data. Khan et al. [63] also applied supervised ML models, including k-nearest neighbors, random forest, and XGBoost, to predict inflation rates and found that these models significantly outperformed conventional benchmarks in terms of precision and forecasting stability. Overall, the integration of supervised ML in economic forecasting offers enhanced model performance, particularly when dealing with complex, high-dimensional datasets where traditional econometric approaches often fall short.

2.3.2 Unsupervised Learning Models

Unsupervised learning techniques are typically used for data preprocessing, feature extraction, or dimensionality reduction, rather than for direct forecasting. One of the most common unsupervised methods in macroeconomic forecasting is Principal Component Analysis (PCA). PCA reduces a large set of economic indicators into a smaller number of uncorrelated principal components, which retain

most of the variation in the original data. These components can then be used as inputs in either traditional or machine learning models [64].

This approach was notably used by Stock and Watson in their work on factor-augmented forecasting models, and has since become a standard technique for reducing the dimensionality of large macroeconomic datasets [65]. PCA is particularly helpful when working with big data, as it reduces multicollinearity and makes model estimation more stable [65].

Other unsupervised methods, such as clustering algorithms (e.g., k-means), have also been explored in the literature [66], [67]. These are often used to group data into distinct economic regimes or periods, such as high vs. low inflation, which can inform the development of regime-specific forecasting models [68]. Although less common than PCA, these methods contribute to structuring macroeconomic data more meaningfully, particularly when used alongside supervised models.

Recent contributions in the literature have demonstrated the practical use of unsupervised learning techniques in macroeconomic forecasting tasks. Xu et al. [69], for instance, proposed a novel global economic policy uncertainty (GEPU) index by integrating PCA with Random Matrix Theory. Their method allowed for a more stable and data-efficient extraction of latent economic signals, which could then be incorporated into forecasting models for improved performance. Other studies have experimented with clustering and anomaly detection techniques to classify economic data into meaningful structures. For example, Kumar et al. [70] applied a multi-source unsupervised transfer learning framework using Isolation Forests to detect macroeconomic pattern shifts from diverse data streams. These approaches, while less common than PCA, provide a valuable preprocessing layer, particularly in environments with high-dimensional or noisy datasets. Although unsupervised methods are not typically used for direct prediction, they play an important role in preparing and structuring data, ultimately improving the effectiveness of both traditional and machine learning forecasting models.

2.3.3 Hybrid and Ensemble Learning Models

Hybrid and ensemble models combine the strengths of multiple forecasting techniques, offering flexibility and robustness in volatile economic environments [71], [72].

A common hybrid setup involves pairing a traditional model, such as ARIMA, with a machine learning model. In this configuration, the ARIMA model is used to capture the linear structure of the time series, while the residuals (i.e., the unexplained part of the model) are passed

to a machine learning model such as an LSTM or XGBoost to model the nonlinear component. This two-stage approach has shown strong performance in forecasting key macroeconomic indicators across several studies [73].

Ensemble methods, such as bagging, boosting, and stacking, are also widely used. These techniques aggregate the predictions of several models to reduce forecasting errors [74]. For instance, XGBoost is itself a form of boosting and has proven highly effective in macroeconomic forecasting contexts. Researchers have also experimented with combining forecasts from traditional and ML models, such as averaging predictions from a VAR and an LSTM model, resulting in better forecast accuracy than either model could achieve on its own [75].

The literature consistently shows that ensemble models perform particularly well in forecasting during periods of high economic uncertainty, where single-model approaches may struggle to generalize. These findings suggest that hybrid and ensemble methods offer a practical pathway to improving the accuracy and stability of macroeconomic forecasts.

Several recent studies further reinforce the value of hybrid and ensemble models in forecasting economic indicators. Zhen et al. [76] proposed a hybrid model that combines multiple machine learning algorithms, including support vector regression and decision trees, to enhance financial time-series forecasting. Their approach showed that combining models at different stages of the forecasting process produced more stable and accurate predictions, particularly in noisy data environments. Similarly, Xavier et al. [77] designed a hybrid inflation forecasting model by integrating Exponential Smoothing with Random Forests, showing that the hybrid configuration outperformed individual models in both short-term and long-term inflation predictions. These findings suggest that hybrid frameworks can effectively balance the interpretability of traditional models with the flexibility of ML techniques.

The effectiveness of ensemble methods is also well-documented in the literature. Patil et al. [78] developed an ensemble model that aggregated multiple machine learning algorithms to predict stock market volatility, concluding that the integrated model consistently outperformed single-model benchmarks. Likewise, Yang et al. [79] applied ensemble learning to improve forecasts of foreign trade flows, demonstrating better generalization and lower forecasting error rates than those produced by traditional models alone. These ensemble strategies, whether through boosting, bagging, or stacking, prove especially valuable in periods of economic uncertainty or when working with complex, high-dimensional datasets.

Overall, the growing body of evidence confirms that hybrid and ensemble learning models are not only theoretically promising but also empirically robust. Their adaptability, combined with their ability to capture both linear and nonlinear patterns, makes them particularly well-suited for forecasting macroeconomic indicators in dynamic and data-rich environments.

2.4 Comparing ML and Traditional Forecasting Models

An important question that arises recently in the literature is how ML-based forecasts compare to forecasts produced by traditional models. Generally, the literature concludes that no single method consistently outperforms across all scenarios, as each forecasting method has its own specific strengths and weaknesses. Additionally, the performance of each approach depends on many factors that include the number of available observations and the economic context. However, several common patterns emerge in the literature.

2.4.1 Advantages of ML Models

One of the most frequently cited strengths of ML models is their ability to handle nonlinear relationships and complex interactions. It is important to highlight that in the context of this study, macroeconomic dynamics often exhibit nonlinear behaviour and complex interconnections between the variables [80]. Therefore, ML models such as neural networks are better equipped to handle such complex interconnections compared to linear models [81].

In addition to the two advantages mentioned above, ML models also have the ability of incorporating large numbers of predictors, which helps reduce the risk of overfitting [61], [82]. This is particularly beneficial during periods of economic stress, when many indicators (e.g., GDP, unemployment, exports, government spending) may jointly signal a downturn [83]. Finally, ML models handle outliers and structural breaks better than traditional methods, which makes them dynamic and quickly adapt to new patterns [83].

2.4.2 Limitations of ML Models

Despite the advantages cited above, ML techniques have some limitations in the context of macroeconomic forecasting. First, while traditional models produce interpretable coefficients (e.g., the impact of a 1% increase in GDP growth), ML models are often considered as “black boxes”, making it difficult to understand how inputs influence outputs [84], [85]. This issue is particularly problematic in macroeconomic forecasting, where policy-making environments require transparency and interpretability. The second limitation is the risk of overfitting, especially when sample sizes are small, which is a common issue in macroeconomic datasets [86]. Finally, ML models

tend to be computationally intensive, requiring more resources and expertise compared to traditional models [60].

2.5 Forecast Evaluation Metrics and Validation Techniques

Regardless of the forecasting method, a rigorous evaluation of forecast accuracy is essential. In the existing literature, various metrics for measuring forecasting errors are used to either assess the model performance or compare two (or more) models [87].

The first and most common accuracy metric used when forecasting continuous variables is the Mean Absolute Error (MAE). The MAE has the same unit as the original data and is less sensitive to outliers compared to the metrics that will be discussed after. This metric is mathematically expressed as [87]:

$$MAE = \frac{\sum_{i=1}^n |y_i - x_i|}{n}$$

Where:

- y_i is the prediction at time i
- x_i is the actual observation at time i
- n is the total number of observations

The second most common error metric, it is the Root Mean Square Error (RMSE) [87]. This metric measures the square root of the average squared differences between predicted and actual values, and can be mathematically expressed as [87]:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2}$$

Where the notations remain the same as previously defined.

Unlike the MAE, the RMSE penalizes large errors, making it more sensitive to outliers.

The third most common metric is the Mean Absolute Percentage Error (MAPE), which is widely used to compare forecasts across different scales [87]. The MAPE measures forecast errors as a percentage of their actual values, allowing for intuitive interpretations. However, this metric becomes problematic when actual values are close to zero. The MAPE can be expressed as [87]:

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - x_i}{x_i} \right|$$

The R-squared is the fourth commonly used metric to assess forecast performance. Unlike the previous metrics that focus on error terms, the R-squared measures the proportion of variance in the actual values that is explained by the model [88]. However, the R-squared should be interpreted with caution, especially for non-linear time series or complex models where it may be misleading.

Finally, in more recent studies, scholars that deal with comparing the performance of different forecasting models use statistical tests such as the Diebold-Mariano test to check if differences in forecast accuracy between two models are significant and not due to chance [89], [90].

It is important to note that many contributions that forecast economic indicators use more than one forecasting technique and suggest the most suitable method based on the lowest forecast error values. For instance, Jafari-Samimi et al. [91] compare between the exponential smoothing and neural network methods to forecast the Iranian GDP. Based on the MAPE, RMSE, and Mean Absolute Deviation (MAD), the authors found that recurring neural networks provided better estimates compared to those produced by the exponential smoothing model. Similarly, the contribution of Alomani et al. [34] used ARIMA and advanced ML to forecast inflation. Based on MAE, Mean Squared Error (MSE), and RMSE, the authors found that ML-based forecasts outperformed those generated by the ARIMA model. Empirical evidence comparing both ML and traditional models is mixed. However, this comparison is generally favourable towards ML in many settings.

In summary, the existing literature shows that macroeconomic forecasting has evolved from relying solely on traditional statistical models such as ARIMA, VAR, and GARCH to the incorporation of machine learning approaches, including supervised, unsupervised, and hybrid techniques. While traditional models remain widely used due to their interpretability and ease of use, they often struggle with complex and nonlinear dynamics that characterize modern economic systems. In contrast, machine learning models, particularly those using neural networks, tree-based algorithms, and ensemble methods, have shown significant promise in improving forecasting accuracy, especially when dealing with high-dimensional data or rapidly changing economic conditions. Unsupervised techniques also play a crucial role by organizing and preparing economic data in ways that enhance the performance of predictive models. Overall, the literature supports a growing consensus that combining the strengths of

traditional and machine learning approaches can lead to more accurate and robust macroeconomic forecasts.

The next section will present the forecasting models selected for this study and describe how they will be applied to predict key macroeconomic indicators.

3 Methodology

This section outlines the methodological framework employed in this study. It begins by presenting the overarching research approach and rationale for model selection. It then details the four forecasting models used—two traditional (ARIMA, VAR) and two machine learning-based (LSTM, XGBoost)—before describing the experimental workflow, including data acquisition, model implementation, performance evaluation, and data processing.

3.1 Research Approach

This study adopts a comparative machine learning approach to evaluate the effectiveness of different models in predicting macroeconomic indicators. The focus is on forecasting key macroeconomic variables using a structured methodology that encompasses data collection, preprocessing, model implementation, and performance evaluation. The overall process followed a data-centric, iterative research methodology. Rather than adhering to a rigid software development model such as Waterfall or Agile, the project evolved through cycles of model implementation, validation, and refinement—each informed by diagnostic results and performance metrics. This flexible approach ensured methodological rigour while enabling adaptive decision-making throughout the analysis.

3.2 Selected Models

To enable a robust and meaningful comparison, this study adopts a diverse set of forecasting models encompassing both traditional econometric approaches and modern machine learning techniques. Specifically, two classical models—ARIMA and VAR—were included for their established roles in time-series and multivariate economic forecasting. Alongside these, two machine learning models—LSTM and XGBoost—were selected to evaluate the potential of data-driven methods in capturing nonlinearities and complex temporal relationships.

These models were chosen to reflect a broad methodological spectrum: univariate versus multivariate, statistical versus algorithmic, and interpretable versus adaptive. This balanced selection allows for a comprehensive assessment of model suitability under varying macroeconomic conditions and supports the study's objective of identifying effective forecasting strategies for GDP and related indicators.

3.3 Workflow

Each model has been trained and evaluated using the same dataset, and results were compared to identify the most effective forecasting

approach. The research methodology is structured into three key phases as outlined below.

3.3.1 Data Acquisition and Preprocessing

- Data has been collected from the Office for National Statistics (ONS). ONS provides rigorously verified and government-approved economic data, ensuring high accuracy, credibility, and consistency over time—critical for forecasting macroeconomic trends.
- The dataset includes key macroeconomic indicators such as GDP, Unemployment Rate, Inflation Rate, Current Account Balance, and Trade Data (Imports and Exports).
- Missing values, scaling issues, and outliers have been handled to prepare the data for modelling.

3.3.2 Model Implementation

- The dataset has been chronologically split into 70% training, 15% validation, and 15% testing sets to preserve temporal dependencies and prevent information leakage.
- Traditional models (VAR, ARIMA) have used only training and testing sets, as their parameters were selected using statistical criteria like Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).
- Machine learning models (LSTM, XGBoost) have been trained using all three subsets, with the validation set used for hyperparameter tuning and early stopping.
- Time series-specific transformations, including lag feature creation and differencing, have been applied where required to ensure stationarity.
- Each model has been tuned using appropriate hyperparameter optimization techniques, such as GridSearchCV or automated methods, depending on the model type.
- All models have been evaluated using a common test set to ensure consistent and fair comparison.

3.3.3 Performance Evaluation

Forecast accuracy has been measured with key performance metrics:

- Mean Absolute Error (MAE): Measures the average magnitude of errors.
- Root Mean Squared Error (RMSE): Penalizes larger errors more than MAE.
- R-squared (R^2): Evaluates how well the model explains variance in macroeconomic indicators.

These metrics have been selected due to their widespread use in both statistical and machine learning literature for time series forecasting. Alternative metrics like Mean Absolute Percentage Error (MAPE) were considered but excluded due to their limitations with zero or near-zero values (common in macroeconomic data such as trade balance or inflation changes). Similarly, Adjusted R^2 is more suited to regression models with multiple independent variables, whereas our models primarily focus on temporal dependencies. A comparative analysis has been conducted to identify the most reliable and efficient model for economic forecasting.

3.4 Data Processing Approach

Preprocessing is a critical step in ensuring the dataset is clean, structured, and suitable for machine learning algorithms. This process has included handling missing values and transforming data for optimal performance.

3.4.1 Handling Missing Values

Missing data can introduce bias and reduce the predictive performance of models. To address this, the dataset has been carefully examined for missing values. For time series data, missing entries have been handled using forward or backward filling methods—commonly used techniques that preserve the temporal structure by propagating the most recent or upcoming known value, respectively.

3.4.2 Data Normalization and Transformation

Since macroeconomic indicators vary significantly in scale (e.g., Unemployment rate, Inflation rate), normalization has been essential to ensure uniformity in data representation.

Min-Max Scaling: To ensure uniformity and prevent dominance of larger numerical values, Min-Max Scaling has been applied:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Where X' is the normalized value, and $X_{min} - X_{max}$ are the feature's minimum and maximum. This transformation rescales variables to $[0,1]$, improving model convergence.

4 Implementation

This section details end-to-end implementation process, including data sourcing, predictor selection, modelling techniques, tools used, and individual model configurations

4.1 Data Sources

The dataset comprises macroeconomic indicators obtained from publicly available financial and economic databases. The primary attributes include:

- **GDP:** Measures economic growth and output.
- **Unemployment Rate:** Indicates labour market conditions.
- **Inflation Rate:** Reflects price stability.
- **Current Account Balance:** Represents trade and financial transactions.
- **Value of Imports and Exports:** Shows trade activity and economic openness.

The data used in this study is sourced from:

- **Office for National Statistics (ONS):** Provides official UK economic data, including GDP, unemployment rate, inflation, and trade balances.

4.2 Macroeconomic Indicators as Predictors

The dataset includes both global and local macroeconomic indicators used as predictors for economic forecasting.

4.2.1 Global Economic Factors

- **Value of Imports/Exports of Goods and Services:** Capture international trade flows and global economic linkages.

4.2.2 Local Economic Factors

- **Gross Domestic Product (GDP):** Measures the total economic output of the country.
- **Unemployment Rate:** Reflects labour market conditions and overall economic stability.
- **Inflation Rate:** Indicates domestic purchasing power and cost-of-living trends.
- **Current Account Balance:** Represents the trade balance and financial health of the national economy.

This refined set of macroeconomic indicators ensures a comprehensive approach to economic forecasting using machine learning techniques.

4.3 Models Implementation

This section outlines the tools, technologies, and processes used to implement the forecasting models, including data preparation, model training, and optimization strategies across ARIMA, VAR, XGBoost, and LSTM approaches.

4.4 Tools and Programming Languages

All forecasting models in this study were developed using Python due to its extensive support for statistical analysis, machine learning, and time-series forecasting. The development and experimentation were primarily conducted on Google Colab, which provided a flexible cloud-based environment. GPU acceleration (specifically Nvidia Tesla T4) was occasionally used within Colab for deep learning tasks like LSTM to speed up training. A MacBook Pro equipped with M2 chip was used for local testing, file organization, and preliminary analysis.

The implementation relied on a variety of well-established Python libraries and frameworks:

- **Pandas, NumPy** – for data manipulation, transformation, and preprocessing
- **Matplotlib, seaborn** – for data visualization and exploratory analysis
- **Statsmodels** – for statistical modelling, including ARIMA and VAR models
- **Xgboost** – for implementing Extreme Gradient Boosting model
- **TensorFlow/Keras** – for building and training LSTM models

Additionally, Microsoft Excel played a key role during the data collection and cleaning phase, allowing for quick manual adjustments and alignment of macroeconomic indicators before there were imported into Python for analysis.

4.5 Individual Model Implementation

This section presents a detailed breakdown of the implementation process for each forecasting model. While the preceding subsections outlined general strategies for data preparation and optimization, the following parts focus on model-specific configurations, preprocessing steps, tuning procedures, and diagnostics carried out.

4.5.1 VAR Model

The VAR model was selected to capture the interdependencies among multiple macroeconomic indicators and assess how past values of all variables jointly influence GDP growth. This multivariate approach is well-suited for time-series datasets with co-evolving variables.

To ensure the validity of VAR model, each time series was tested for stationarity using the Augmented Dickey-Fuller (ADF) test. As shown in Table 1, only the GDP series was stationary in levels. All other variables required first-order differencing.

Variable	ADF Statistic	p-value	Stationary
GDP	-4.4069	0.003	Yes
Inflation	-1.7509	0.4051	No
Unemployment	-2.7670	0.0631	No
Current Account	-2.5036	0.1146	No
Exports	-0.6828	0.8512	No
Imports	-0.0205	0.9568	No

Table 1: ADF Test Results Before Differencing

To determine the optimal lag order for the VAR model, a grid search was conducted evaluating performance across lags 1 to 8 using forecasting metrics (MAE, RMSE, and R^2). And the best performance was observed at lag 2, which was used for the final model.

To assess residual behaviour, diagnostic plots were generated for the GDP residuals from the fitted VAR(2) model. As shown in Figure 1 and Figure 2, the residuals appear randomly distributed around zero and the autocorrelation function (ACF) shows no significant lags, suggesting the residuals resemble white noise.

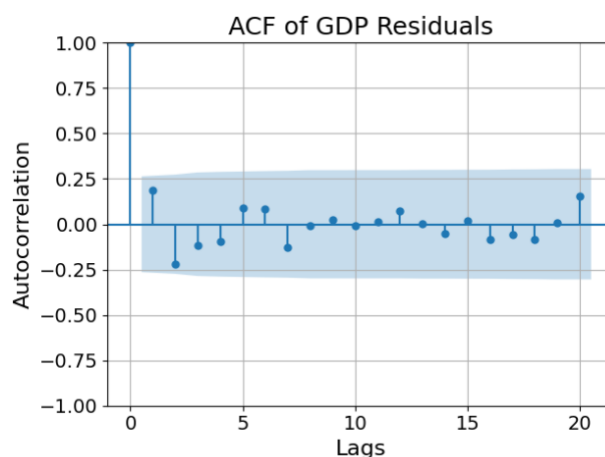


Figure 1: ACF of GDP Residuals

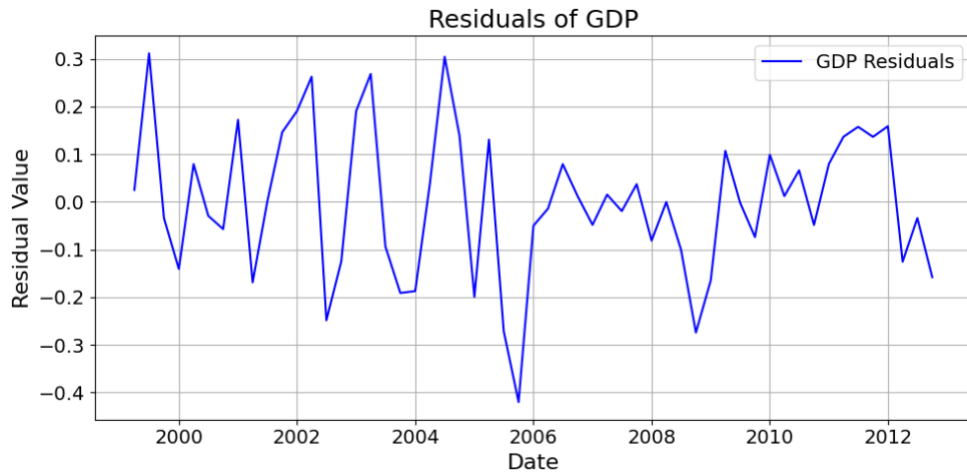


Figure 2: Time Series Plot of GDP Residuals from the VAR(2) Model

4.5.2 ARIMA Model

The ARIMA model was implemented to forecast GDP growth using only past GDP values. Prior the model fitting, the stationarity of GDP series was tested using the ADF test. As shown in Table 2, the test statistic was -4.4069 with a p-value of 0.0003, indicating strong evidence against the null hypothesis of non-stationary. Therefore, no differencing was required, and the original series was used.

Test Statistic	p-value	1% CV	5% CV	10%CV
-4.4069	0.0003	-3.506	-2.895	-2.584

Table 2: ADF Test Results for GDP

To identify the optimal ARIMA configuration, a grid search was conducted across various combinations of the (p,d,q) parameters. The top three models based on the AIC is shown in Table 3. The ARIMA(3,0,0) model was selected based on the lowest AIC score.

Order	AIC	BIC
(3,0,0)	97.79	110.40
(3,1,1)	98.99	111.55
(3,0,1)	99.72	114.85

Table 3: Top 3 ARIMA Models by AIC

Residual analysis was conducted to validate the model. As shown in Figure 3, the residuals appear randomly distributed around zero, with no apparent trend or pattern. In addition, the autocorrelation of the residuals was assessed using ACF and PACF plots, displayed in Figure 4, which showed no significant lags, further supporting the adequacy of the model.

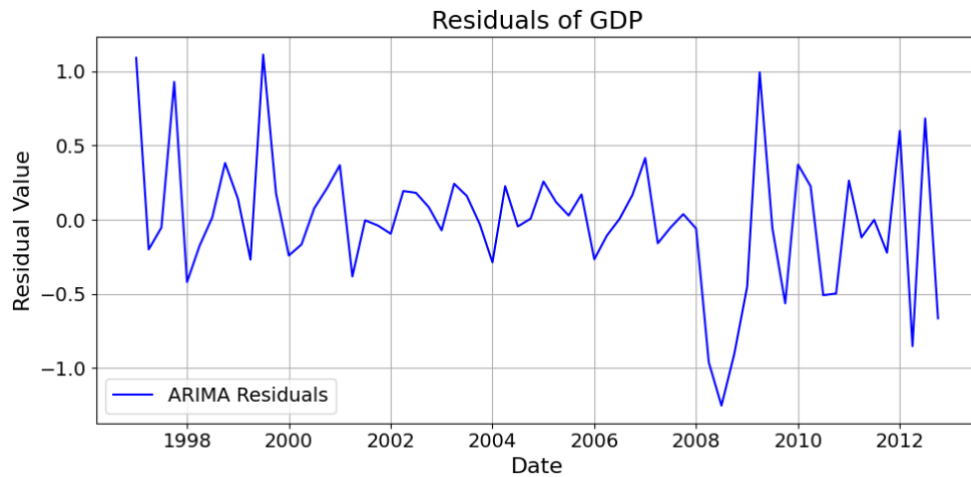


Figure 3: Residual Plot of ARIMA(3,0,0)

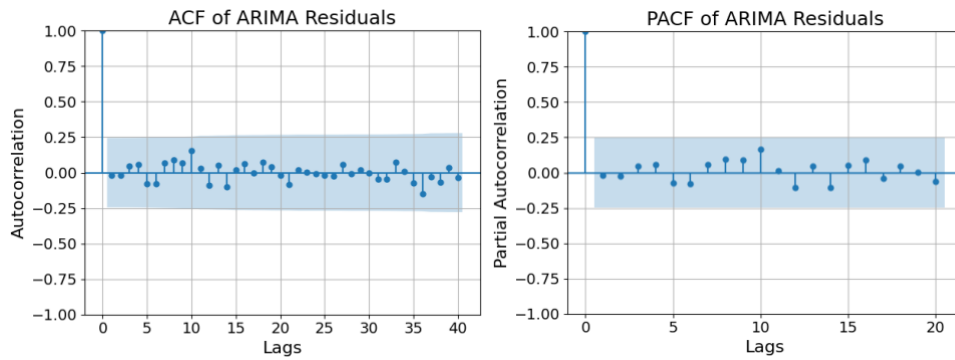


Figure 4: ACF and PACF of Residuals for ARIMA(3,0,0)

4.5.3 LSTM Model

The LSTM model was implemented to capture nonlinear temporal dependencies in GDP growth. As a univariate time-series model, it was trained exclusively on the GDP series, using a lag of 4 quarters (one year) to construct the input sequences.

The data was first normalised using MinMax scaling, then split chronologically into training (70%), validation (15%), and testing (15%) sets. Each input sample consisted of four consecutive quarters of GDP values used to predict the following quarter.

The model architecture consisted of a single LSTM layer with 50 units and ReLU activation, followed by a dense output layer. The network was compiled with the Adam optimizer and mean squared error as the loss function. Training was performed over 100 epochs with a batch size of 4, using the validation set to monitor performance during learning.

Figure 5 presents the training and validation loss over epochs, indicating stable convergence without overfitting.

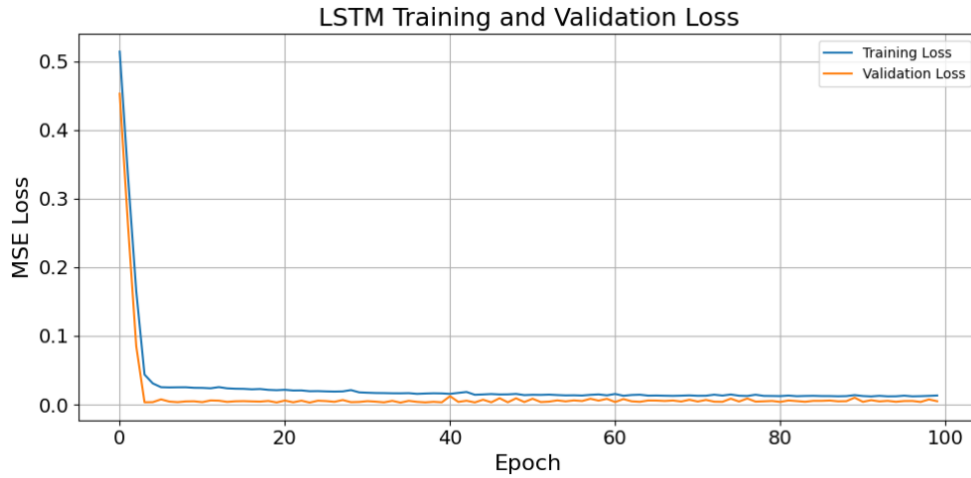


Figure 5: LSTM Training and Validation Loss Across Epochs

Due to its recurrent structure, the LSTM model did not require manual feature engineering or differencing for stationary, unlike ARIMA and VAR. The sequential data structure and internal memory mechanisms allowed it to learn temporal patterns directly from the raw GDP sequence.

Residual diagnostics such as ACF and PACF were reviewed separately, as shown in Figure 6, to confirm the independence of residuals and absence of autocorrelation.

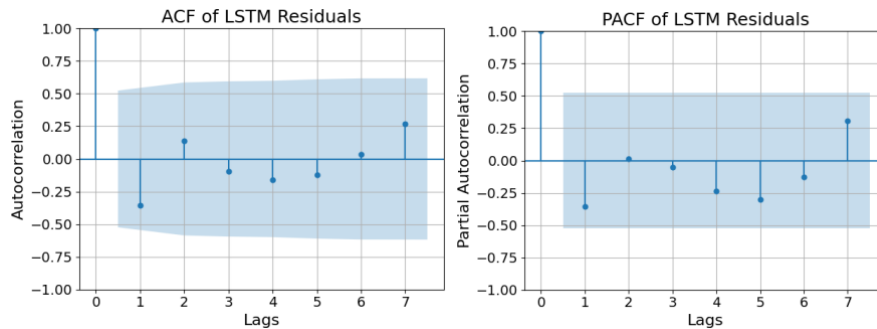


Figure 6: ACF and PACF of LSTM Residual

4.5.4 XGBoost Model

The XGBoost model was designed as a supervised learning regression task using tabular data constructed from macroeconomic indicators. Unlike time-series models, XGBoost does not require sequential input and instead benefits from feature engineering that captures temporal dynamics.

Lagged values of GDP and other variables were used to provide the model with historical context. Additional features such as moving averages, rate-of-change indicators, trade ratios, and interaction terms (e.g., Inflation x Unemployment) were included to enrich the input space and allow the model to learn complex relationships.

The dataset was split chronologically into 70% training, 15% validation, and 15% testing subsets. All input features were scaled using the MinMaxScaler to optimize model performance. A grid of manually tuned hyperparameters was used during training, including a shallow tree depth (`max_depth=2`), slow learning rate (`learning_rate=0.005`), and regularization terms (`reg_alpha`, `reg_lambda`) to minimize overfitting.

The final was evaluated on the test set, and the top contributing features were visualised to interpret which macroeconomic factors had the greatest influence on GDP prediction, as shown in Figure 6.

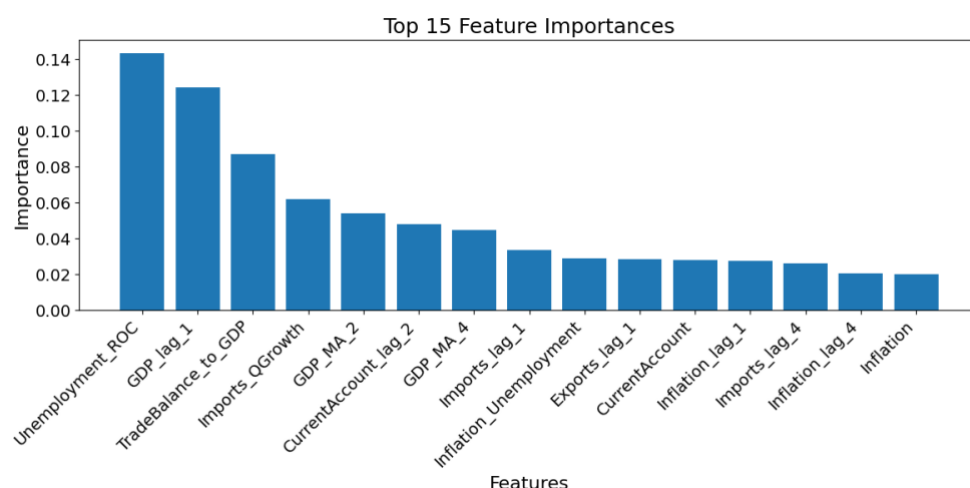


Figure 7: Top 15 Feature Importances in XGBoost GDP Forecasting

5 Results and Evaluation

This section presents the forecasting performance of each model applied to GDP growth prediction. Evaluation metrics include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . Each model is first analysed individually using its forecast results on the test set. A final subsection compares all models, synthesising outcomes into a comprehensive performance summary.

5.1 VAR Model Evaluation

The VAR model was evaluated using a lag order of 2, selected through grid search by minimising RMSE across validation folds. Trained on a differenced multivariate dataset including inflation, unemployment, trade balance, and current account indicators, the model aimed to exploit potential interdependencies to improve GDP growth forecasting.

Despite leveraging multiple economic indicators, the model's performance was limited. It yielded an MAE of 0.2562, an RMSE of 0.2991, and an R^2 of -0.5162. The negative R^2 highlights that the model performed worse than a simple mean-based prediction, failing to generalise effectively to unseen data. In practical terms, the model explained no meaningful proportion of variance in GDP growth, undermining its reliability for forecasting under volatile conditions.

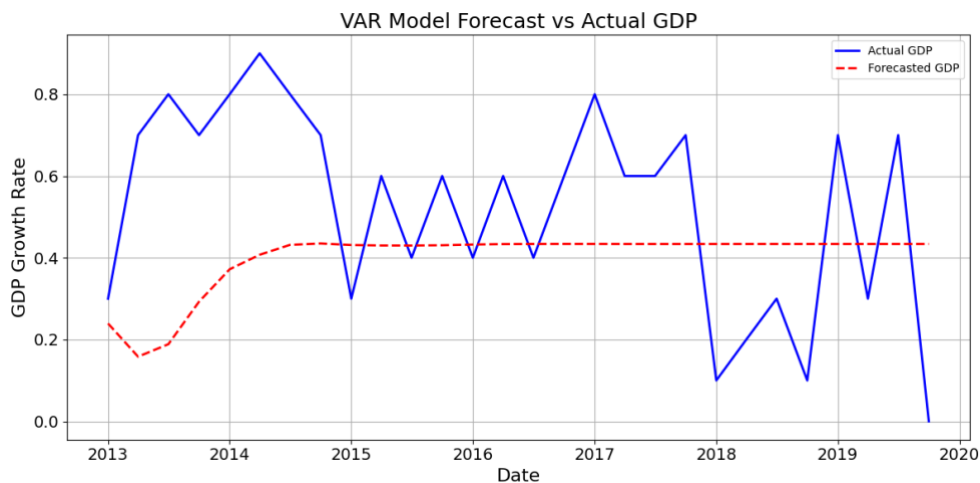


Figure 8: Var Model Forecast vs Actual GDP

As shown in Figure 8, the predicted values follow an overly smooth trajectory, failing to respond to the sharp fluctuations present in the actual GDP growth series. This is a typical limitation of linear models like VAR when applied to economic systems characterised by nonlinear shifts and structural breaks. Additionally, the flatness of the forecast line suggests that the model's responsiveness was

dampened by differencing and multivariate noise, which may have diluted relevant predictive signals.

These results indicate that while VAR is theoretically well-suited for modelling macroeconomic interdependencies, in practice, its effectiveness is heavily dependent on data quality, stationarity transformations, and careful lag structure selection—all of which were addressed in this study but yielded limited gains.

5.2 ARIMA Model Evaluation

The ARIMA model, configured with the order (3,0,0) via grid search optimization, demonstrated moderate improvements over the VAR model in terms of forecast precision. It achieved an MAE of 0.2337, an RMSE of 0.2717, and an R^2 of -0.2508. Although R^2 remained negative—indicating underperformance relative to a naive mean forecast—the reduction in MAE and RMSE suggests better alignment with short-term GDP fluctuations.

Figure 9 shows that the ARIMA forecast initially diverges from GDP, dipping while the actual series jumps, then tracks the upward drift with reduced volatility and eventually turns flat. By smoothing over sharp swings, the model responds slowly to rapid economic changes and understates extreme movements in the data. Even so, ARIMA provides a stable, interpretable baseline against which richer models can be compared.

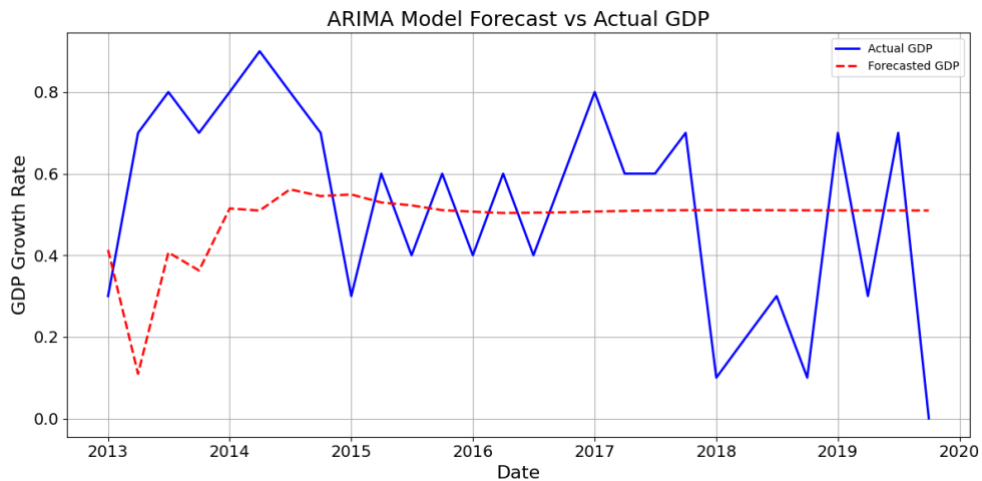


Figure 9: ARIMA Model Forecast vs Actual GDP

5.3 LSTM Model Evaluation

The LSTM model outperformed both ARIMA and VAR in terms of forecasting accuracy, achieving the lowest MAE of 0.2236 and RMSE

of 0.2673. Although the R^2 value remained negative at -0.0729, it was significantly closer to zero, indicating a smaller deviation from actual values compared to previous models.

Unlike the traditional models, LSTM captured nonlinear and temporal dependencies more effectively. As shown in Figure 10, its predictions display a more adaptive trajectory, aligning more closely with the changing direction of GDP growth, particularly during inflection points. While some variation and amplitude mismatches are still visible, the LSTM curve adjusts better to rises and drops in the test period than its counterparts, reflecting its ability to learn complex sequential patterns without explicit feature engineering.

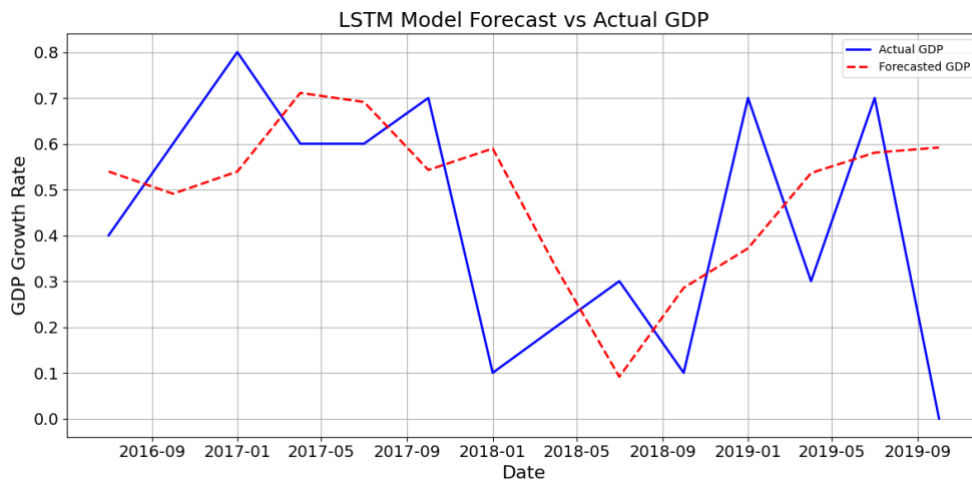


Figure 10: LSTM Model Forecast vs Actual GDP

Despite the dataset's limited size, the LSTM's recurrent structure contributed to relatively smoother and responsive forecasts. This suggests that with larger datasets or multivariate inputs, its performance could potentially improve further.

5.4 XGBoost Model Evaluation

The XGBoost model achieved the strongest overall performance among all models tested. It recorded an MAE of 0.1969, RMSE of 0.2176, and the only positive R^2 value of 0.3386, indicating that it was able to explain a meaningful proportion of the variance in the GDP test set—unlike the other models which yielded negative R^2 scores.

Compared to more traditional forecasting approaches, XGBoost demonstrated a superior capacity to generalize patterns in the data, capturing key movements and directional shifts in GDP more reliably. This improvement is largely attributable to its ability to exploit engineered lag features and nonlinear interactions without requiring sequential input.

As illustrated in Figure 11, the forecasted GDP line closely follows the actual GDP trajectory, with notably fewer large deviations than seen in previous models. Although minor discrepancies remain—particularly in capturing sudden spikes or drops—the overall predictive pattern aligns well with real economic movements. This performance highlights XGBoost’s advantage when working with well-structured tabular datasets enhanced through feature engineering, and its utility for high-stakes economic forecasting applications.

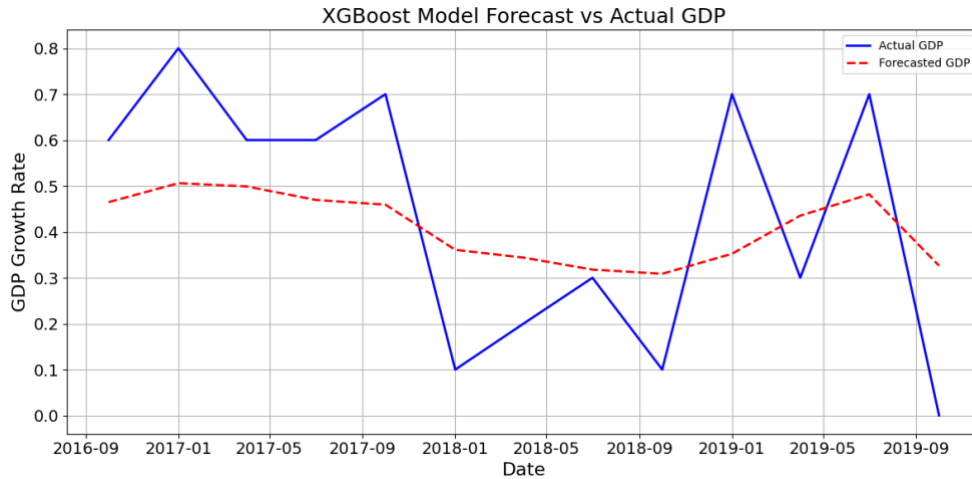


Figure 11: XGBoost Forecast vs Actual GDP

5.5 Models Comparison

This section presents a comparative analysis of the forecasting performance of all implemented models—VAR, ARIMA, LSTM, and XGBoost—based on MAE, RMSE, and R^2 . These metrics are standard for evaluating model accuracy and explanatory power. The results are summarised in Table 4.

Model	MAE	RMSE	R^2
VAR	0.2562	0.2991	-0.5162
ARIMA	0.2337	0.2717	-0.2508
LSTM	0.2236	0.2673	-0.0729
XGBoost	0.1969	0.2176	0.3386

Table 4: Evaluation metrics for each model on the dataset.

As shown in Table 4, XGBoost delivered the strongest performance across all three metrics. With the lowest MAE and RMSE, and an R^2 of 0.3386, it was the only model that explained a meaningful portion of the variance in GDP growth. This confirms the suitability of tree-based machine learning models for capturing nonlinear relationships in macroeconomic data.

LSTM followed closely, with competitive MAE and RMSE values, and an R^2 approaching zero (-0.0729), suggesting it nearly matched a naive prediction baseline. Its recurrent architecture allowed it to learn temporal dependencies effectively, despite a relatively small dataset.

ARIMA showed slightly inferior performance, with a modest reduction in error metrics over VAR, but still a negative R^2 (-0.2508). Its simplicity and focus on univariate series may have limited its ability to fully adapt to dynamic GDP patterns.

VAR, though theoretically capable of leveraging multiple macroeconomic indicators, produced the highest error values and the most negative R^2 (-0.5162), indicating substantial overfitting or inability to generalise from training to test data.

These results collectively underscore the importance of model selection in time series forecasting. While traditional models like ARIMA and VAR provide interpretability and solid baselines, more sophisticated machine learning approaches—particularly XGBoost—offered significantly improved accuracy, making them valuable tools for economic forecasting tasks that involve complex, nonlinear trends.

A visual comparison of the predicted vs actual GDP trajectories is shown in Figure 12, highlighting the differences in forecast accuracy and trend tracking across models.

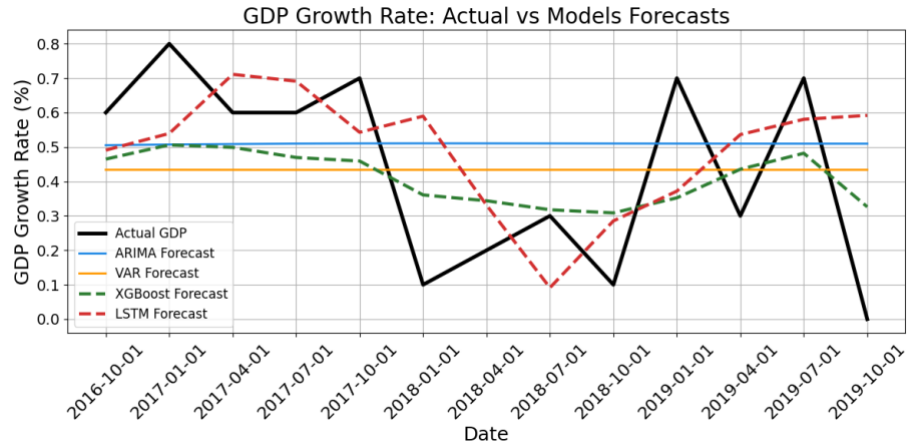


Figure 12: GDP Growth Rate: Actual vs Models Forecasts

6 Conclusion

This project set out to investigate the comparative performance of classical and ML models in forecasting UK GDP using macroeconomic indicators; this section reflects on the overall findings, their implications, technical complexities encountered and potential directions for future work and improvements.

This study compared the forecasting performance of four models—VAR, ARIMA, LSTM, and XGBoost—for predicting UK GDP growth. XGBoost consistently delivered the highest accuracy across key evaluation metrics (MAE, RMSE, and R^2), confirming its strength in modeling nonlinear relationships and handling structured macroeconomic features through advanced feature engineering.

The LSTM model, while slightly behind XGBoost in performance, still demonstrated a notable advantage over classical models. Its recurrent architecture enabled it to learn temporal patterns without requiring explicit feature design. However, the implementation posed challenges including selecting an appropriate sequence window (set to 4 quarters), configuring hyperparameters (units, batch size, learning rate), and managing training stability. These steps required careful tuning and validation to avoid overfitting, particularly due to the limited size of the dataset.

The ARIMA model served as a reliable baseline, benefiting from interpretability and statistical rigor. Rather than using Auto-ARIMA, a full grid search was conducted to identify optimal parameters, which proved computationally demanding. A particular challenge arose in ensuring consistent model comparison: ARIMA required stationary inputs, whereas the deep learning and ensemble models did not, necessitating tailored preprocessing strategies to align performance evaluations.

VAR, despite its multivariate formulation, yielded the weakest performance. Although its structure allowed for modelling interactions among multiple indicators, it was highly sensitive to the selection of lag orders and to noise from less informative variables. Selecting the optimal lag length involved exhaustive search and evaluation across multiple configurations.

Beyond the modelling challenges, considerable effort went into assembling and curating the dataset. Many macroeconomic indicators had differing start dates, requiring all variables to be realigned to a common time frame. This process involved sacrificing a portion of potentially useful data in order to preserve indicator consistency. Additionally, decisions had to be made about the trade-off between including more indicators versus having a longer historical time

window—choices that impacted model complexity, reliability, and generalizability.

Taken together, the outcomes support the project's hypothesis that advanced machine learning models, particularly tree-based ensembles like XGBoost, can significantly enhance macroeconomic forecasting accuracy when paired with thoughtful feature design and preprocessing. At the same time, the project illustrated the nuanced decisions and technical constraints that shape the practical application of forecasting methods.

Future Work

While this study has demonstrated the viability of multiple forecasting approaches for GDP prediction, several areas remain open for improvement and future exploration.

- **Multivariate LSTM or Transformer-based Models:** Future experiments could explore multivariate deep learning architectures that incorporate multiple macroeconomic indicators rather than using GDP alone.
- **Model Ensembling:** Combining forecasts from different models (e.g., ARIMA + XGBoost or LSTM + VAR) could leverage the strengths of each and improve predictive robustness.
- **Feature Importance and Explainability:** Expanding interpretability tools such as SHAP values for XGBoost or attention mechanisms in LSTM could enhance understanding of model decisions.
- **Longer Forecast Horizons:** This study focused on short-term forecasting; extending to multi-step or long-term predictions may reveal new insights and challenges.
- **External Shocks and Exogenous Variables:** Including external shocks (e.g., pandemic effects, policy changes) or exogenous variables through models like ARIMAX or dynamic VAR could provide richer forecasting capabilities.

These directions can help build more accurate, interpretable, and adaptive models for economic forecasting in future research.

Collectively, these findings highlight the value of combining traditional economic theory with modern machine learning methods. By bridging these approaches, the project contributes to a more adaptable and data-driven foundation for future economic forecasting efforts.

Appendices

Appendix A: ADF Test Results After Differencing (VAR)

Variable	ADF Statistic	P-value	Stationary
GDP	-4.8639	0.0000	Yes
Inflation	-4.9187	0.0000	Yes
Unemployment	-3.8626	0.0023	Yes
Current Account	-5.5933	0.0000	Yes
Exports	-7.3780	0.0000	Yes
Imports	-6.0863	0.0000	Yes

Table 5: ADF Test Results After Differencing

Appendix B: ARIMA Model Selection Based on BIC

Order	BIC	AIC
(1,0,0)	110.11	102.54
(3,0,0)	110.40	97.80
(3,1,1)	111.55	98.99

Table 6: Top 3 ARIMA Models by BIC

Appendix C: VAR Lag Selection Grid Search

	Lag	MAE	RMSE	R2
1	2	0.256243	0.299136	-0.516190
0	1	0.266367	0.307940	-0.606753
2	3	0.282156	0.352454	-1.104855
4	5	0.320292	0.405000	-1.779244
3	4	0.372633	0.519992	-3.581528
5	6	1.415873	1.661736	-45.788676
6	7	2.664703	3.274786	-180.711517
7	8	36.749652	59.738167	-60466.268240

Figure 13: Grid Search Results for VAR Model

Appendix D: XGB Model Configuration

```
# Define parameters directly
params = {
    'objective': 'reg:squarederror',
    'learning_rate': 0.005,
    'max_depth': 2,
    'min_child_weight': 2,
    'subsample': 0.7,          # More aggressive subsampling
    'colsample_bytree': 0.8,
    'n_estimators': 500,       # More trees with slower learning rate
    'reg_alpha': 0.1,          # L1 regularization
    'reg_lambda': 1.0,         # L2 regularization
    'random_state': 42
}

# Train model
model = xgb.XGBRegressor(**params)
```

Figure 14: Final Hyperparameter configuration used for the XGB Model

Appendix E: LSTM Model Structure and Training

```
# Build the LSTM model
model = Sequential()
model.add(LSTM(50, activation='relu', input_shape=(SEQ_LEN, 1)))
model.add(Dense(1))
model.compile(optimizer='adam', loss='mse')

# Train
history = model.fit(X_train, y_train, epochs=100, batch_size=4,
                    validation_data=(X_val, y_val), verbose=0)
```

Figure 15: Architecture and Training Configuration of the LSTM Model

Bibliography

- [1] R. Veeramachaneni, “Predictive forecasting in economics: The imperfect crystal ball,” *International Journal of Computer Engineering and Technology (IJCET)*, vol. 15, no. 5, Art. no. 5, Sep. 2024.
- [2] J. A. Ogilvy, *Creating better futures: Scenario planning as a tool for a better tomorrow*. Oxford University Press, USA, 2002.
- [3] G. Mathy and H. Stekler, “Expectations and forecasting during the Great Depression: Real-time evidence from the business press,” *Journal of Macroeconomics*, vol. 53, pp. 1–15, Sep. 2017, doi: 10.1016/j.jmacro.2017.05.006.
- [4] C. Heij, D. van Dijk, and P. J. F. Groenen, “Real-time macroeconomic forecasting with leading indicators: An empirical comparison,” *International Journal of Forecasting*, vol. 27, no. 2, pp. 466–481, Apr. 2011, doi: 10.1016/j.ijforecast.2010.04.008.
- [5] S. C. Agu, F. U. Onu, U. K. Ezemagu, and D. Oden, “Predicting gross domestic product to macroeconomic indicators,” *Intelligent Systems with Applications*, vol. 14, p. 200082, May 2022, doi: 10.1016/j.iswa.2022.200082.
- [6] E. I. Sukhanova, S. Y. Shirnaeva, and A. G. Mokronosov, “Econometric models for forecasting of macroeconomic indices,” *International Journal of Environmental and Science Education*, vol. 11, no. 16, pp. 9191–9205, 2016.
- [7] M. E. Pérez-Pons, J. Parra-Dominguez, S. Omatu, E. Herrera-Viedma, and J. M. Corchado, “Machine learning and traditional econometric models: A systematic mapping study,” *Journal of Artificial Intelligence and Soft Computing Research*, vol. 12, no. 2, pp. 79–100, Feb. 2022, doi: 10.2478/jaiscr-2022-0006.

- [8] P. Goulet Coulombe, M. Leroux, D. Stevanovic, and S. Surprenant, "How is machine learning useful for macroeconomic forecasting?," *Journal of Applied Econometrics*, vol. 37, no. 5, pp. 920–964, 2022, doi: 10.1002/jae.2910.
- [9] K. Dynan and L. Sheiner, "GDP as a measure of economic well-being," in *The Measure of Economies: Measuring Productivity in an Age of Technological Change*, M. B. Reinsdorf and L. Sheiner, Eds., University of Chicago Press, 2024, pp. 9–50. doi: 10.7208/chicago/9780226836348-002.
- [10] T. Modis, "Long-term GDP forecasts and the prospects for growth," *Technological Forecasting and Social Change*, vol. 80, no. 8, pp. 1557–1562, Oct. 2013, doi: 10.1016/j.techfore.2013.02.010.
- [11] G. Tkacz, "Neural network forecasting of Canadian GDP growth," *International Journal of Forecasting*, vol. 17, no. 1, pp. 57–69, Jan. 2001, doi: 10.1016/S0169-2070(00)00063-7.
- [12] E. F. Mumbuli, E. N. Abelly, M. M. Mgimba, and A. E. Twum, "Towards precision economics: Unveiling GDP patterns using integrated deep learning techniques," *Comput Econ*, Jan. 2025, doi: 10.1007/s10614-025-10863-x.
- [13] T. Callen, "Back to basics - What is gross domestic product?," *Finance and development: F&D; a quarterly publication of The International Monetary Fund*, vol. 45, no. 4, 2008.
- [14] M. V. der Steen, "Chapter 11: Anticipation tools in policy formulation: forecasting, foresight and implications for policy planning," 2017. Accessed: Mar. 23, 2025. [Online]. Available: <https://www.elgaronline.com/edcollchap/edcoll/9781784719319/9781784719319.00018.xml>

- [15] R. Zhang and H. Ji, “GDP forecast of Shanghai based on Grey Model,” *Academic Journal of Business & Management*, vol. 3, no. 9, Nov. 2021, doi: 10.25236/AJBM.2021.030906.
- [16] S. Moshiri and N. Cameron, “Neural network versus econometric models in forecasting inflation,” *Journal of Forecasting*, vol. 19, no. 3, pp. 201–217, 2000, doi: 10.1002/(SICI)1099-131X(200004)19:3<201::AID-FOR753>3.0.CO;2-4.
- [17] J. H. Stock and M. W. Watson, “Forecasting inflation,” *Journal of Monetary Economics*, vol. 44, no. 2, pp. 293–335, Oct. 1999, doi: 10.1016/S0304-3932(99)00027-6.
- [18] R. O’Neill, J. Ralph, and P. A. Smith, “What is inflation?,” in *Inflation: History and Measurement*, R. O’Neill, J. Ralph, and P. A. Smith, Eds., Cham: Springer International Publishing, 2017, pp. 21–43. doi: 10.1007/978-3-319-64125-6_2.
- [19] D. Borup and E. C. M. and Schütte, “In search of a job: Forecasting employment growth using Google Trends,” *Journal of Business & Economic Statistics*, vol. 40, no. 1, pp. 186–200, Jan. 2022, doi: 10.1080/07350015.2020.1791133.
- [20] D. E. Rapach and J. K. Strauss, “Forecasting US employment growth using forecast combining methods,” *Journal of Forecasting*, vol. 27, no. 1, pp. 75–93, 2008, doi: 10.1002/for.1051.
- [21] G. Duffee, “Chapter 7 - Forecasting interest rates,” in *Handbook of Economic Forecasting*, vol. 2, G. Elliott and A. Timmermann, Eds., in *Handbook of Economic Forecasting*, vol. 2. , Elsevier, 2013, pp. 385–426. doi: 10.1016/B978-0-444-53683-9.00007-4.
- [22] J. Beckmann and R. Schüssler, “Forecasting exchange rates under parameter and model uncertainty,” *Journal of International Money and*

Finance, vol. 60, pp. 267–288, Feb. 2016, doi: 10.1016/j.jimonfin.2015.07.001.

[23] C.-M. Kuan and T. Liu, “Forecasting exchange rates using feedforward and recurrent neural networks,” *Journal of Applied Econometrics*, vol. 10, no. 4, pp. 347–364, 1995, doi: 10.1002/jae.3950100403.

[24] A. Pescatori and S. Zaman, “Macroeconomic models, forecasting, and policymaking,” *Economic Commentary*, no. 2011–19, Oct. 2011, doi: 10.26509/frbc-ec-201119.

[25] P. Beaudry and T. Willems, “On the macroeconomic consequences of over-optimism,” *American Economic Journal: Macroeconomics*, vol. 14, no. 1, pp. 38–59, 2022.

[26] L. Qin, K. Shanks, G. Phillips, and D. Bernard, “The impact of lengths of time series on the accuracy of the ARIMA forecasting,” *International Research in Higher Education*, vol. 4, p. 58, Aug. 2019, doi: 10.5430/irhe.v4n3p58.

[27] P. Mondal, L. Shit, and S. Goswami, “Study of effectiveness of time series modeling (ARIMA) in forecasting stock prices,” *International Journal of Computer Science, Engineering and Applications*, vol. 4, pp. 13–29, Apr. 2014, doi: 10.5121/ijcsea.2014.4202.

[28] S. L. Ho and M. Xie, “The use of ARIMA models for reliability forecasting and analysis,” *Computers & Industrial Engineering*, vol. 35, no. 1, pp. 213–216, Oct. 1998, doi: 10.1016/S0360-8352(98)00066-7.

[29] J. Fattah, L. Ezzine, Z. Aman, H. El Moussami, and A. Lachhab, “Forecasting of demand using ARIMA model,” *International Journal of Engineering Business Management*, vol. 10, p. 1847979018808673, Jan. 2018, doi: 10.1177/1847979018808673.

- [30] A.-C. Petrică, S. Stancu, and A. Tindeche, “Limitation of ARIMA models in financial and monetary economics,” *Theoretical and Applied Economics*, vol. XXIII, no. 4(609), Winter, pp. 19–42, 2016.
- [31] A. Vafin, “Forecasting macroeconomic indicators for seven major economies using the ARIMA model,” vol. 3, no. 1, p. 1, 2020, doi: 10.17613/j3b4-h115.
- [32] Y. E. Kuzu and S. Alp, “Estimating the macroeconomic indicators using ARIMA and ANFIS methods,” *Recent Advances in Science and Engineering*, vol. 2, no. 1, p. 6, 2022, doi: 10.14744/rase.2022.0002.
- [33] K. M. S. Uddin and N. Tanzim, “Forecasting GDP of Bangladesh using ARIMA model,” *International Journal of Business and Management*, vol. 16, no. 6, Art. no. 6, May 2021, doi: 10.5539/ijbm.v16n6p56.
- [34] G. Alomani, M. Kayid, and M. F. Abd El-Aal, “Global inflation forecasting and Uncertainty Assessment: Comparing ARIMA with advanced machine learning,” *Journal of Radiation Research and Applied Sciences*, vol. 18, no. 2, p. 101402, Jun. 2025, doi: 10.1016/j.jrras.2025.101402.
- [35] B. Advani, A. Sachan, U. K. Sahu, and A. K. Pradhan, “Forecasting major macroeconomic variables of the Indian economy,” in *Modeling Economic Growth in Contemporary India*, B. S. Sergi, A. K. Tiwari, and S. Nasreen, Eds., Emerald Publishing Limited, 2024, pp. 1–24. doi: 10.1108/978-1-80382-751-320241001.
- [36] Y. Ma, “Analysis and forecasting of GDP using the ARIMA model,” *Information Systems and Economics*, vol. 5, no. 1, pp. 91–97, Feb. 2024, doi: 10.23977/infse.2024.050112.
- [37] N. Fokin and A. Polbin, “Forecasting Russia’s key macroeconomic indicators with the VAR-LASSO model,” *Russian Journal of Money and Finance*, vol. 78, pp. 67–93, Jun. 2019, doi: 10.31477/rjmf.201902.67.

- [38] A. D'Agostino, L. Gambetti, and D. Giannone, "Macroeconomic forecasting and structural change," *Journal of Applied Econometrics*, vol. 28, no. 1, pp. 82–101, 2013, doi: 10.1002/jae.1257.
- [39] M. Rockinger and G. and Urga, "A time varying parameter model to test for predictability and integration in the stock markets of transition economies," *Journal of Business & Economic Statistics*, vol. 19, no. 1, pp. 73–84, Jan. 2001, doi: 10.1198/07350010152472634.
- [40] A. Wilhelmsson, "GARCH forecasting performance under different distribution assumptions," *Journal of Forecasting*, vol. 25, no. 8, pp. 561–578, 2006, doi: 10.1002/for.1009.
- [41] D. Barbara, C. Li, Y. Jing, and A. Samuel, "Modeling and forecast of Ghana's GDP using ARIMA-GARCH model," *Open Access Library Journal*, vol. 9, no. 1, Art. no. 1, Jan. 2022, doi: 10.4236/oalib.1108335.
- [42] R. J. Abreu, R. M. Souza, and J. G. Oliveira, "Applying singular spectrum analysis and ARIMA-GARCH for forecasting EUR/USD exchange rate," *RAM, Rev. Adm. Mackenzie*, vol. 20, p. eRAMF190146, Aug. 2019, doi: <https://doi.org/10.1590/1678-6971/eRAMF190146>.
- [43] C. Uwilingiyimana, J. tu, and J. Harerimana, "Forecasting inflation in Kenya using ARIMA-GARCH models," *International Journal of Management and Commerce Innovations*, vol. 3, pp. 2348–758515, Nov. 2015.
- [44] M. Omane-Adjepong, F. Oduro, and S. Oduro, "Determining the better approach for short-term forecasting of Ghana's inflation: Seasonal ARIMA Vs Holt-Winters," *International Journal of Business, Humanities and Technology*, vol. 3, pp. 69–79, Jan. 2013.
- [45] D. Gabrielyan, "Forecasting inflation using the Phillips curve in inflation targeting countries," *International Review of Applied Economics*,

vol. 33, no. 5, pp. 601–623, Sep. 2019, doi: 10.1080/02692171.2018.1516740.

[46] T. Iyer and A. S. Gupta, “Quarterly forecasting model for India’s economic growth: Bayesian vector autoregression approach,” ADB Economics Working Paper Series, Working Paper 573, 2019. doi: 10.22617/WPS190056-2.

[47] G. Rumbe, M. Hamasha, and S. Al Mashaqbeh, “A comparison of Holts-Winter and Artificial Neural Network approach in forecasting: A case study for tent manufacturing industry,” *Results in Engineering*, vol. 21, p. 101899, Mar. 2024, doi: 10.1016/j.rineng.2024.101899.

[48] C. Chatfield and M. Yar, “Holt-Winters forecasting: Some practical issues,” *Journal of the Royal Statistical Society Series D: The Statistician*, vol. 37, no. 2, pp. 129–140, Jun. 1988, doi: 10.2307/2348687.

[49] A. G. Sleeman, “Retrospectives: The Phillips Curve: A rushed job?,” *Journal of Economic Perspectives*, vol. 25, no. 1, pp. 223–238, Mar. 2011, doi: 10.1257/jep.25.1.223.

[50] F. B. Santos and T. R. C. Stekel, “The impact of economic indicators on consumption: A neural network approach,” *International Contemporary Management Review*, vol. 5, no. 3, pp. e215–e215, Dec. 2024, doi: 10.54033/icmr5n3-043.

[51] E. Umronov, A. Kadirov, A. Abdujabborov, I. Muydinov, M. Karimova, and T. Sobirjonov, “Economic levels forecasting system by evaluating with more accuracy using ML, DL and AI systems,” in *2024 4th International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)*, May 2024, pp. 1027–1031. doi: 10.1109/ICACITE60783.2024.10616869.

[52] G. Pillonetto, T. Chen, A. Chiuso, G. De Nicolao, and L. Ljung, “Regularization of linear regression model,” in *Regularized System*

Identification: Learning Dynamic Models from Data, G. Pillonetto, T. Chen, A. Chiuso, G. De Nicolao, and L. Ljung, Eds., Cham: Springer International Publishing, 2022, pp. 33–93. doi: 10.1007/978-3-030-95860-2_3.

[53] F. Emmert-Streib and M. Dehmer, “High-dimensional LASSO-based computational regression models: Regularization, shrinkage, and selection,” *Machine Learning and Knowledge Extraction*, vol. 1, no. 1, Art. no. 1, Mar. 2019, doi: 10.3390/make1010021.

[54] X. Li, “Smartphone price prediction using decision tree and support vector regression (SVR),” *Applied and Computational Engineering*, vol. 115, pp. 37–43, Dec. 2024, doi: 10.54254/2755-2721/2025.18475.

[55] A. Gifty and D. Y. Li, “A comparative analysis of LSTM, ARIMA, XGBoost algorithms in predicting stock price direction,” *Engineering And Technology Journal*, vol. 9, no. 8, pp. 4978–4986, Aug. 2024, doi: 10.47191/etj/v9i08.50.

[56] J. Zhang, “Enhancing predictive models in E-commerce: A comparative study using XGBoost across diverse scenarios,” *ITM Web Conf.*, vol. 70, p. 02014, 2025, doi: 10.1051/itmconf/20257002014.

[57] S. Qureshi, B. M. Chu, and F. S. Demers, “Forecasting Canadian GDP growth using XGBoost,” *Carleton Economic Papers*, Art. no. 20–14, Aug. 2020, Accessed: Mar. 23, 2025. [Online]. Available: <https://ideas.repec.org/p/car/carecp/20-14.html>

[58] B. Rai, “Long short-term memory (LSTM) deep neural networks for sentiment classification,” in *Artificial Intelligence for Signal Processing and Wireless Communication*, A. Sharma, A. Jain, A. Kumar Arya, and M. Ram, Eds., De Gruyter, 2022, pp. 1–10. doi: 10.1515/9783110734652-001.

[59] J. P. Tan, A. L. A. Ramos, M. V. Abante, R. L. Tadeo, and R. R. Lansigan, “A performance review of recurrent neural networks long short-term memory (LSTM),” in *2022 3rd International Conference for Emerging*

Technology (INCET), May 2022, pp. 1–5. doi: 10.1109/INCET54531.2022.9824567.

[60] B. Oancea and M. Simionescu, “Gross domestic product forecasting: Harnessing machine learning for accurate economic predictions in a univariate setting,” *Electronics*, vol. 13, no. 24, Art. no. 24, Jan. 2024, doi: 10.3390/electronics13244918.

[61] C. Zhangyin, “Research on machine learning forecasting model of economic indicators based on big data,” in *2024 International Conference on Electronics and Devices, Computational Science (ICEDCS)*, Sep. 2024, pp. 809–816. doi: 10.1109/ICEDCS64328.2024.00151.

[62] A. Gupta, K. Kumar, V. Asha, and S. N. Kumar, “Machine learning in economic forecasting: A comprehensive review,” in *2024 7th International Conference on Contemporary Computing and Informatics (IC3I)*, Sep. 2024, pp. 882–886. doi: 10.1109/IC3I61595.2024.10828960.

[63] M. A. Khan *et al.*, “Application of machine learning algorithms for sustainable business management based on macro-economic data: Supervised learning techniques approach,” *Sustainability*, vol. 14, no. 16, Art. no. 16, Jan. 2022, doi: 10.3390/su14169964.

[64] J. Ma, “Progress and application of unsupervised feature extraction methods,” *Applied and Computational Engineering*, vol. 106, pp. 99–104, Nov. 2024, doi: 10.54254/2755-2721/106/20241313.

[65] J. H. Stock and M. W. and Watson, “Forecasting using principal components from a large number of predictors,” *Journal of the American Statistical Association*, vol. 97, no. 460, pp. 1167–1179, Dec. 2002, doi: 10.1198/016214502388618960.

[66] G. Aoki, K. Ataka, T. Doi, and K. Tsubouchi, “Data-driven estimation of economic indicators with search big data in discontinuous situation,” *The*

Journal of Finance and Data Science, vol. 9, p. 100106, Nov. 2023, doi: 10.1016/j.jfds.2023.100106.

[67] E. A. Costa and M. E. Silva, “Predicting macroeconomic indicators from online activity data: A review,” *Statistical Journal of the IAOS*, vol. 40, no. 2, pp. 403–419, May 2024, doi: 10.3233/SJI-230102.

[68] S. Q. Noor and T. K. Mustafa, “Comparing K-Means, Nearest Neighbor, and Lloyd’s Clustering Algorithms,” *Iraqi Journal of Science*, pp. 6688–6704, Nov. 2024, doi: 10.24996/ij.s.2024.65.11.40.

[69] W. Xu, W. Rao, L. Wei, and Q. Wang, “A normalized global economic policy uncertainty index from unsupervised machine learning,” *Mathematics*, vol. 11, no. 15, Art. no. 15, Jan. 2023, doi: 10.3390/math11153268.

[70] S. Kumar, A. K. Shukla, and P. K. Muhuri, “Isolation forest based multi-source unsupervised transfer learning for missing GDP prediction,” in *2020 International Joint Conference on Neural Networks (IJCNN)*, Jul. 2020, pp. 1–8. doi: 10.1109/IJCNN48605.2020.9207482.

[71] A. Atesongun and M. Gulsen, “A hybrid forecasting structure based on ARIMA and artificial neural network models,” *Applied Sciences*, vol. 14, no. 16, Art. no. 16, Jan. 2024, doi: 10.3390/app14167122.

[72] V. Sharma, M. Rakhra, V. Sharma, T. Sarkar, A. Singh, and K. Jairath, “Hybrid approaches for stocks price prediction: A new institutive way of neural architecture design,” in *2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT)*, Aug. 2024, pp. 1–6. doi: 10.1109/ICEECT61758.2024.10739249.

[73] R. Nichani, L. Gasmi, N. Laiche, and S. Kabou, “Optimizing financial time series predictions with hybrid ARIMA, LSTM, and XGBoost Models,” *STUDIES IN ENGINEERING AND EXACT SCIENCES*, vol. 5, no. 2, pp. e11188–e11188, Nov. 2024, doi: 10.54021/seesv5n2-582.

- [74] C. A. R. Goyzueta, J. E. C. De la Cruz, and E. R. A. Estaño, “Regression ensemble techniques bagging, Gradient Boosting and XGBoost to predict the power of photovoltaic systems in the High Andean Area,” in *2024 IEEE XXXI International Conference on Electronics, Electrical Engineering and Computing (INTERCON)*, Nov. 2024, pp. 1–6. doi: 10.1109/INTERCON63140.2024.10833486.
- [75] D. Patil and K. Alaskar, “Integrating weighted averaging and boosting in ensemble models for improved flood forecasting,” in *2024 International Conference on Intelligent Systems and Advanced Applications (ICISAA)*, Oct. 2024, pp. 1–6. doi: 10.1109/ICISAA62385.2024.10828928.
- [76] H. Zhen, F. Cai, Y. Zhang, Z. Mi, and J. Li, “Research on Financial Data Forecasting Based on Machine Learning under New Productive Forces,” in *2024 6th International Conference on Machine Learning, Big Data and Business Intelligence (MLBDBI)*, Nov. 2024, pp. 115–119. doi: 10.1109/MLBDBI63974.2024.10823851.
- [77] A. L. S. Xavier, B. J. T. Fernandes, and J. F. L. de Oliveira, “Hybrid Model and Ensemble for Inflation Forecasting: A Machine Learning Approach,” in *2023 IEEE Latin American Conference on Computational Intelligence (LA-CCI)*, Oct. 2023, pp. 1–6. doi: 10.1109/LA-CCI58595.2023.10409426.
- [78] B. V. Patil *et al.*, “Integrating ensemble machine learning with technical indicators for superior FX volatility prediction,” in *2024 International Conference on Intelligent Systems and Advanced Applications (ICISAA)*, Oct. 2024, pp. 1–6. doi: 10.1109/ICISAA62385.2024.10828793.
- [79] C.-H. Yang, C.-F. Lee, and P.-Y. Chang, “Export- and import-based economic models for predicting global trade using deep learning,” *Expert Systems with Applications*, vol. 218, p. 119590, May 2023, doi: 10.1016/j.eswa.2023.119590.

- [80] T. Kyriazos and M. Poga, “Application of machine learning models in social sciences: Managing nonlinear relationships,” *Encyclopedia*, vol. 4, no. 4, Art. no. 4, Dec. 2024, doi: 10.3390/encyclopedia4040118.
- [81] P. Ganguly and I. Mukherjee, “Enhancing retail sales forecasting with optimized machine learning models,” in *2024 4th International Conference on Sustainable Expert Systems (ICSES)*, Oct. 2024, pp. 884–889. doi: 10.1109/ICSES63445.2024.10762950.
- [82] Y. O. Silahtaroglu, “Machine learning integration in econometric models,” *Next Generation Journal for The Young Researchers*, vol. 8, no. 1, pp. 77–77, Nov. 2024, doi: 10.62802/8c33p210.
- [83] L. Omolo and N. Nguyen, “Using an ensemble of machine learning algorithms to predict economic recession,” *Journal of Risk and Financial Management*, vol. 17, no. 9, Art. no. 9, Sep. 2024, doi: 10.3390/jrfm17090387.
- [84] M. Guo, Q. Zhang, and D. D. Zeng, “An interpretable deep learning-based model for decision-making through Piecewise Linear Approximation,” *ACM Trans. Knowl. Discov. Data*, vol. 19, no. 3, p. 56:1-56:35, Feb. 2025, doi: 10.1145/3715150.
- [85] G. Amir, S. Bassan, and G. Katz, “Hard to explain: On the computational hardness of in-distribution model interpretation,” in *ECAI 2024*, IOS Press, 2024, pp. 818–825. doi: 10.3233/FAIA240567.
- [86] X.-S. Chen, M. G. Kim, C.-H. Lin, and H. J. Na, “Development of per capita GDP forecasting model using deep learning: Including Consumer Goods Index and unemployment rate,” *Sustainability*, vol. 17, no. 3, Art. no. 3, Jan. 2025, doi: 10.3390/su17030843.
- [87] D. Mehrotra, “Performance comparison of time series data using predictive data mining techniques,” *Advances in Information Mining*, vol. 4, no. 1, pp. 57–66, Jan. 2012.

- [88] D. Chicco, M. J. Warrens, and G. Jurman, "The coefficient of determination R-squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation," *PeerJ Comput. Sci.*, vol. 7, p. e623, Jul. 2021, doi: 10.7717/peerj-cs.623.
- [89] F. X. Diebold, "Comparing predictive accuracy, twenty years later: A personal perspective on the use and abuse of Diebold–Mariano tests," *Journal of Business & Economic Statistics*, vol. 33, no. 1, pp. 1–1, Jan. 2015, doi: 10.1080/07350015.2014.983236.
- [90] E. Gilleland and G. Roux, "A new approach to testing forecast predictive accuracy," *Meteorological Applications*, vol. 22, no. 3, pp. 534–543, 2015, doi: 10.1002/met.1485.
- [91] A. Jafari-Samimi, B. Shirazi, and H. Fazlollahtabar, "A comparison between time series, exponential smoothing, and neural network methods to forecast GDP of Iran," *Iranian Economic Review*, vol. 12, no. 19, pp. 19–35, Dec. 2007, doi: 10.22059/ier.2007.32639.